

FLUNIVET 5% w/v INJECTION

Description and Composition

FLUNIVET 5% w/v INJECTION is a colorless to light yellow clear solution. Each ml contains 50mg Flunixin (as meglumine) as its active ingredient and 5mg Phenol as preservative.

Pharmacodynamics

Flunixin meglumine is a relatively potent non-narcotic, non-steroidal analgesic with anti-inflammatory, anti-endotoxic and anti-pyretic properties.

Flunixin meglumine acts as a reversible inhibitor of cyclo-oxygenase, an important enzyme in the arachidonic acid cascade pathway which is responsible for converting arachidonic acid to cyclic endoperoxides. Consequently, synthesis of eicosanoids, important mediators of the inflammatory process involved in central pyresis, pain perception and tissue inflammation, is inhibited. Through its effects on the arachidonic acid cascade, flunixin also inhibits the production of thromboxane, a potent platelet pro-aggregator and vasoconstrictor which is released during blood clotting. Flunixin exerts its antipyretic effect by inhibiting prostaglandin E₂ synthesis in the hypothalamus. By inhibiting the arachidonic acid cascade pathway, flunixin also produces an anti-endotoxic effect by suppressing eicosanoid formation and therefore preventing their involvement in endotoxin associated disease states.

Pharmacokinetics

Flunixin was administered intravenously to horses as a single dose of 1.1mg/kg. At the first timepoint measured (10 minutes after administration) the plasma concentration was 11.45 µg/ml, AUC was 21.45 µg.h/ml and the elimination half-life was approximately 2 hours.

Flunixin was administered intravenously to cattle as a single dose of 2.2 mg/kg. At the first timepoint measured (10 minutes after administration) the plasma concentration was 12.32 µg/ml, AUC was 14.87 µg.h/ml and the elimination half-life was approximately 4 hours.

In an experimental study, flunixin was administered intravenously to pigs as a single dose of 2.0 mg/kg. Flunixin was >98% protein bound at all physiologically relevant concentrations, but also had a large volume of distribution at steady-state. All plasma concentrations were below the limit of quantitation (0.02 µg/ml) by 48 hours and the elimination half-life was 7.76 hours.

Indication

Cattle: Reduction of acute inflammation associated with respiratory disease. Adjunctive therapy in the treatment of acute mastitis.

Horses: Alleviation of inflammation and pain associated with musculo-skeletal disorders. Alleviation of visceral pain associated with colic. Adjunctive therapy in the treatment of endotoxaemia and septic shock.

Pigs: Adjunctive therapy in the treatment of swine respiratory diseases.

Recommended Dose

For intravenous administration to cattle and horses.

Cattle: The recommended dose rate is 2.2mg Flunixin per kg body weight equivalent to 2ml per 45kg body weight. Repeat as necessary at 24 hour intervals for up to 5 consecutive days.

Horses:

For use in equine colic, the recommended dose rate is 1.1mg Flunixin per kg body weight equivalent to 1ml per 45kg body weight. Treatment may be repeated once or twice if colic recurs.

For use in musculo-skeletal disorders, the recommended dose rate is 1.1mg Flunixin per kg body weight equivalent to 1ml per 45kg body weight, once daily for up to 5 days according to clinical response.

For the treatment of endotoxaemia or septic shock associated with gastric torsion and with other conditions in which the circulation of blood to the gastrointestinal tract is compromised: 0.25mg per kg (1ml per 200kg) every 6 to 8 hours.

For intramuscular injection to pigs.

Pigs: The recommended dose rate is 2ml per 45kg body weight (equivalent to 2.2mg Flunixin per kg) once by intramuscular injection, in the neck, in conjunction with appropriate antimicrobial therapy. The injection volume should be limited to a maximum of 5ml per injection site.

Route of Administration

Cattle and horses: By intravenous injection

Pigs: By intramuscular injection

Contraindication

Use is contraindicated in animals suffering from cardiac, hepatic or renal disease, where there is the possibility of gastrointestinal ulceration or bleeding, where there is evidence of a blood dyscrasia.

Do not use in case of hypersensitivity to flunixin meglumine, other NSAIDs or any of the excipients.

Do not use in case of haemorrhagic disorders.

Do not use in animals suffering from chronic musculo-skeletal disorders.

Do not administer to pregnant sows, gilts at mating and in breeding boars.

Do not use the product within 48 hours before expected parturition in cows.

Warning and Precautions

Special warning for target species:

The cause of the underlying inflammatory condition should be determined and treated with appropriate concomitant therapy.

Special precautions for use:

Special precautions for use in animals

Avoid intra-arterial injection.

Use in any animal less than 6 weeks of age or in aged animals may involve additional risk. If such use cannot be avoided animals may require a reduced dosage and careful clinical management.

Avoid use in any dehydrated, hypovolaemic or hypotensive animal except in the case of endotoxaemia or septic shock. It is preferable that NSAIDs, which inhibit prostaglandin synthesis are not administered to animals undergoing general anaesthesia until fully recovered.

Do not use in piglets weighting less than 6 kg.

NSAIDs are known to have the potential to delay parturition through a tocolytic effect by inhibiting prostaglandins that are important in signalling the initiation of parturition. The use of the product in the immediate postpartum period may interfere with uterine involution and expulsion of foetal membranes resulting in retained placentae.

Special precautions to be taken by the person administering the veterinary medicinal product to animals

The product may cause reactions in sensitive individuals. If you have known hypersensitivity for non-steroidal anti-inflammatory products and/or to polyethylene glycol do not handle the product. Reactions may be serious.

To avoid possible sensitisation reactions, avoid contact with the skin. Gloves should be worn during application.

In case of skin contact, wash exposed area with plenty of water and soap. If symptoms persist seek medical advice.

Avoid eye contact. In the case of accidental contact with eyes, rinse immediately with plenty of water and seek medical advice.

Avoid accidental self-injection. In case of accidental self-injection, seek medical advice immediately and show the package leaflet or the label to the physician.

Wash hands after use.

Interaction with Other Medicinal Product

Monitor drug compatibility closely where adjunctive therapy is required.

Do not administer other non-steroidal anti-inflammatory drugs (NSAIDs) concurrently or within 24 hours of each other, as it may increase the toxicity, mainly gastro-intestinal, even with low doses of acetylsalicylic acid. Some NSAIDs may be highly bound to plasma proteins and compete with other highly bound drugs which can lead to toxic effects.

The concurrent administration of corticoids may increase toxicity of the two products and increase the risk of gastro-intestinal ulceration. It should therefore be avoided.

Flunixin may reduce the effect of some anti-hypertensive medicinal product by inhibition of the prostaglandins synthesis, such as diuretics, Angiotensin Conversion Enzyme (ACE) inhibitors and beta blockers. Concurrent administration of potentially nephrotoxic drugs, particularly aminoglycosides, should be avoided.

Flunixin may reduce renal elimination of some drugs and increase their toxicity (for example, aminoglycosides).

Pregnancy and Lactation

May be used in pregnant and lactating cattle.

For pregnant mares, use only according to the benefit/risk assessment by the Responsible Veterinarian.

Do not administer to pregnant sows, gilts at mating and in breeding boars.

The product should only be administered within the first 36 hours postpartum following a benefit/risk assessment performed by the responsible veterinarian and treated animals should be monitored for retained placentae.

Side Effects

Flunixin meglumine is a non steroidal anti-inflammatory drug (NSAID).

Untoward effects include gastrointestinal irritation, ulceration, hepatic idiosyncratic reactions, and, especially in dehydrated or hypovolaemic animals, potential for renal damage.

In pigs, transient irritation may occur at the injection site, this resolves spontaneously within 14 days.

Symptoms and Treatment of Overdose

Overdosage is associated with gastrointestinal toxicity.

Storage Condition

Store below 30°C.

Shelf life

3 years. Following withdrawal of the first dose, use the product within 24 hours. Any unused material should be discarded.

Withdrawal Periods

Cattle: Meat and offal: 10 days

Horses: Meat and offal: 10 days

Pigs: Meat and offal: 22 days

Milk: 24 Hours

The product is not authorised for use in lactating mares producing milk for human consumption.

Packing

20ml, 50ml, 100ml, 250ml.

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